QUALITY INFORMATION DOCUMENT

For Iberian-Biscay-Ireland Ocean Heat Content (0-2000m)

OMI_CLIMATE_OHC_IBI_area_averaged_anomalies

Issue: 1.1

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2022/06/16	All	Initial version	de Pascual Collar, A. Aznar, R. Amo, A. Sotillo, M.G.	
1.1	2026/03/31	All	Copernicus dataset IBI-INT added to the ensemble Added information published in OSR#7	de Pascual Collar, A. Ciliberti, S.	Copernicus Marine Product Management

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1) INTRODUCTION

a) Product Name

OMI_CLIMATE_OHC_IBI_area_averaged_anomalies

b) Indicator Evaluation Method

The Iberia-Biscay-Ireland (IBI) Ocean Heat Content (OHC) indicator provides annually averaged, spatially averaged time series of OHC for the IBI region (26°N–56°N, 19°W–5°E), along with the associated error bars.

The error bars of the product are derived from the computation of the same parameter using a set of gridded Copernicus products available for the region (see Section 1.c).

For each specific ensemble product, OHC is computed following the methodology described in Lima et al. (2020) and Mayer et al. (2021), using the following equation:

$$OHC = \int_{z_1}^{z_2} \rho_0 c_p (T_{mth} - T_{clim}) dz$$

where T_{clim} corresponds to the average temperature over entire data record, including all available temporal record, T_{mth} corresponds to the temperature averaged for a specific month.
A reference density of ρ_0 = 1030 kg·m^{-3} and a specific heat capacity of c_p = 3980 J/kg°C were used (e.g. von Schuckmann et al., 2009). This variable is closely proportional to the average temperature change from z1 = 0 m to z2 = 2000 m depth.

c) Copernicus Marine products used as input for the computation

Product reference ID, type and a cronim	Documentation
IBI_MULTIYEAR_PHY_005_002 Reanalysis **IBI-MYP**	• **PUM:** https://documentation.marine.copernicus.eu/PUM/CMEMS-IBI-PUM-005-002.pdf • **QUID:** https://documentation.marine.copernicus.eu/QUID/CMEMS-IBI-QUID-005-002.pdf
IBI_MULTIYEAR_PHY_005_002 Interim **IBI-INT**	
GLOBAL_MULTIYEAR_PHY_001_030 Reanalysis **GLO-MYP**	• **PUM:** https://documentation.marine.copernicus.eu/PUM/CMEMS-GLO-PUM-001-030.pdf • **QUID:** https://documentation.marine.copernicus.eu/QUID/CMEMS-GLO-QUID-001-030.pdf
INSITU_GLO_PHY_TS_OA_MY_013_052 Observations **CORA**	• **PUM:** https://documentation.marine.copernicus.eu/PUM/CMEMS-INS-PUM-013-002-052.pdf • **QUID**: https://documentation.marine.copernicus.eu/QUID/CMEMS-INS-QUID-013-052.pdf
MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012 Reprocessed observations **ARMOR**	• **PUM:** https://documentation.marine.copernicus.eu/PUM/CMEMS-MOB-PUM-015-012.pdf • **QUID:** https://documentation.marine.copernicus.eu/QUID/CMEMS-MOB-QUID-015-012.pdf

2) SCIENTIFIC VALIDATION

The quality evaluation of IBI_OMI_OHC_area_averaged_anomalies is based on the "multi-product" approach as introduced in the second issue of the Ocean State Report (von Schuckmann et al., 2018) following the MyOcean's experience (Masina et al., 2017). Three global products and a regional (Iberia-Biscay-Ireland) product have been used to build an ensemble mean and its associated ensemble spread (Figure 1, black line and shaded area).

Results of Fig. 1 indicate how each ensemble member is distributed around the ensemble mean and show a continuous increase of OHC over the period 1993-2024 at a rate of 0.55±0.3 W/m^2 in the upper 2000m.

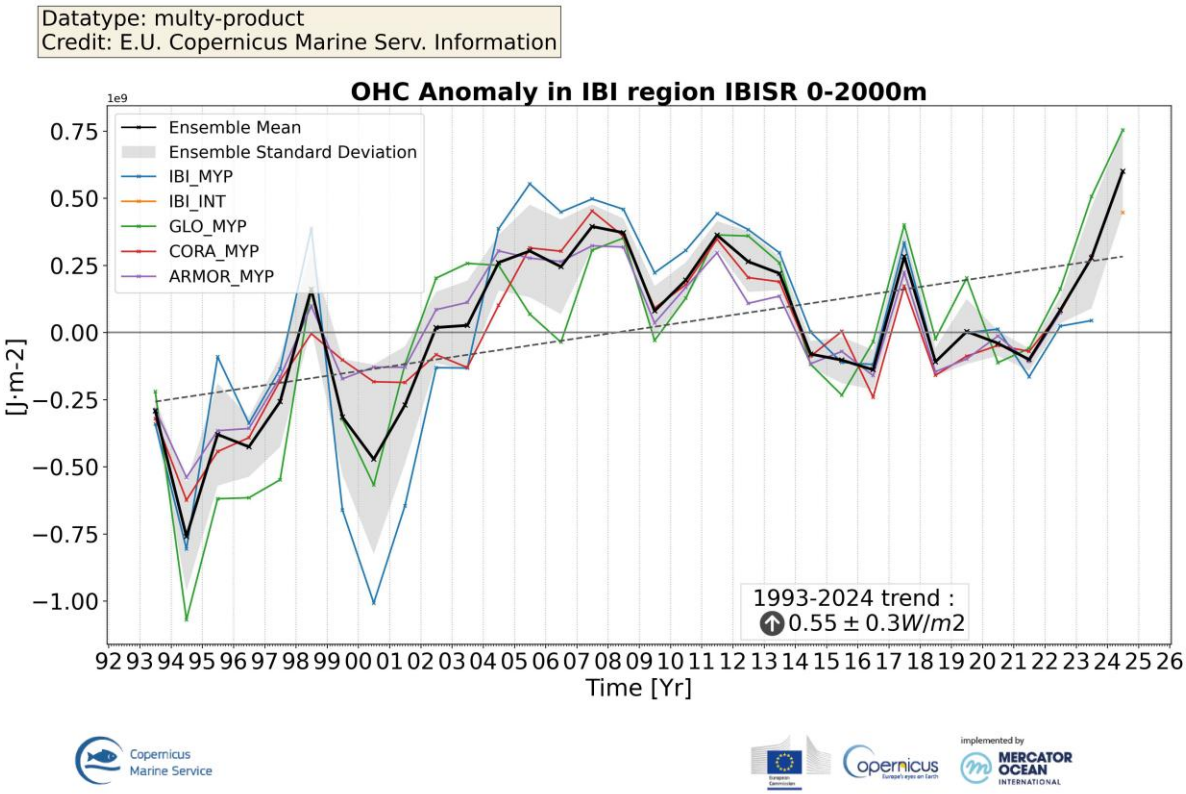

Figure 1: *Time series of area averaged ocean heat content in the Iberia-Biscay-Ireland region and integrated over the 0-2000m depth layer. Time series are based on different data products, i.e. IBI-MYP (blue), IBI-INT (orange), GLO-MYP (green), CORA (red), ARMOR (purple), and the ensemble mean (black) and the ensemble spread (grey shaded).*

More details and full scientific evaluation can be found in the CMEMS Ocean State Report (von Schuckmann et al., 2016; von Schuckmann et al., 2018).

Subsequent studies conducted on this indicator (de Pascual Collar et al., 2023) show that, although its variability and results are consistent and reliable, the high interannual variability of the resulting time series causes the derived trends—while statistically representative—to depend considerably on the selected integration depth (see Figure 2) and on the chosen time period (see Figure 3).

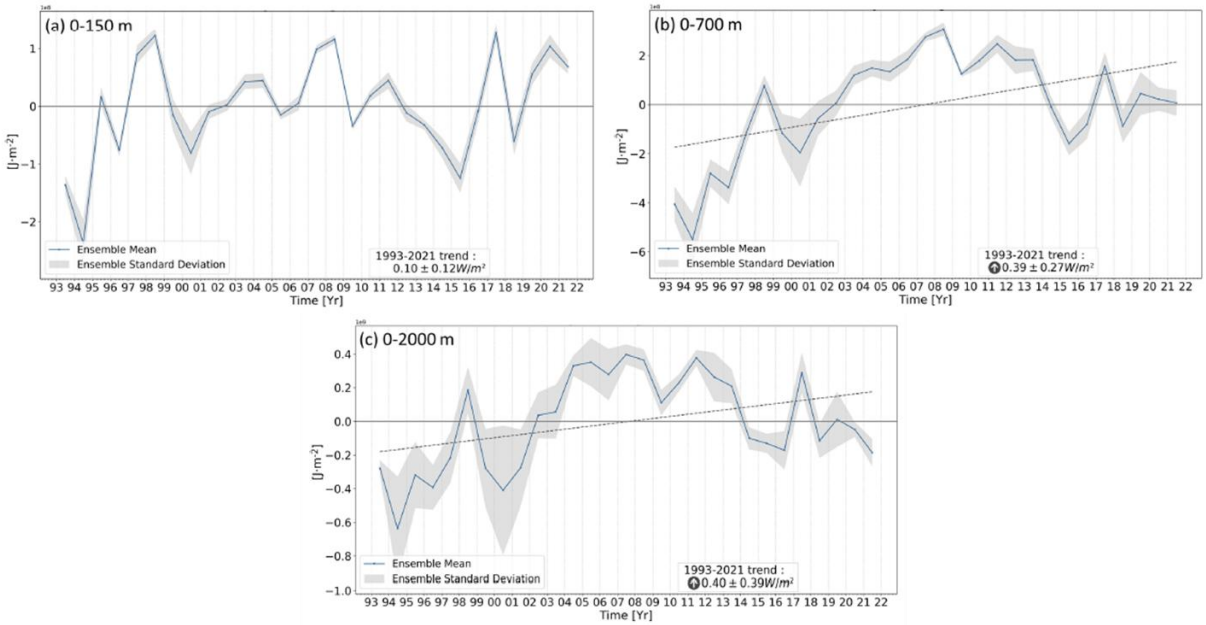

Figure 2: OHC averaged in the IBI domain integrated from the ocean's surface down to 150m (a), 700m (b), and 2000m (c). Time series computed from four Copernicus Marine products (i.e. IBIREA, GLO-REA, CORA, and ARMOR), providing a multi-product approach. The blue line represents the ensemble mean, and shaded grey areas represent the standard deviation of the ensemble. The analysis of trends (at 95% confidence interval) computed in the period 1993–2021 is included (bottom right box). Trend lines (dashed line) are only included in the figures when a significant trend is obtained (figure obtained from de Pascual Collar et al., 2023)

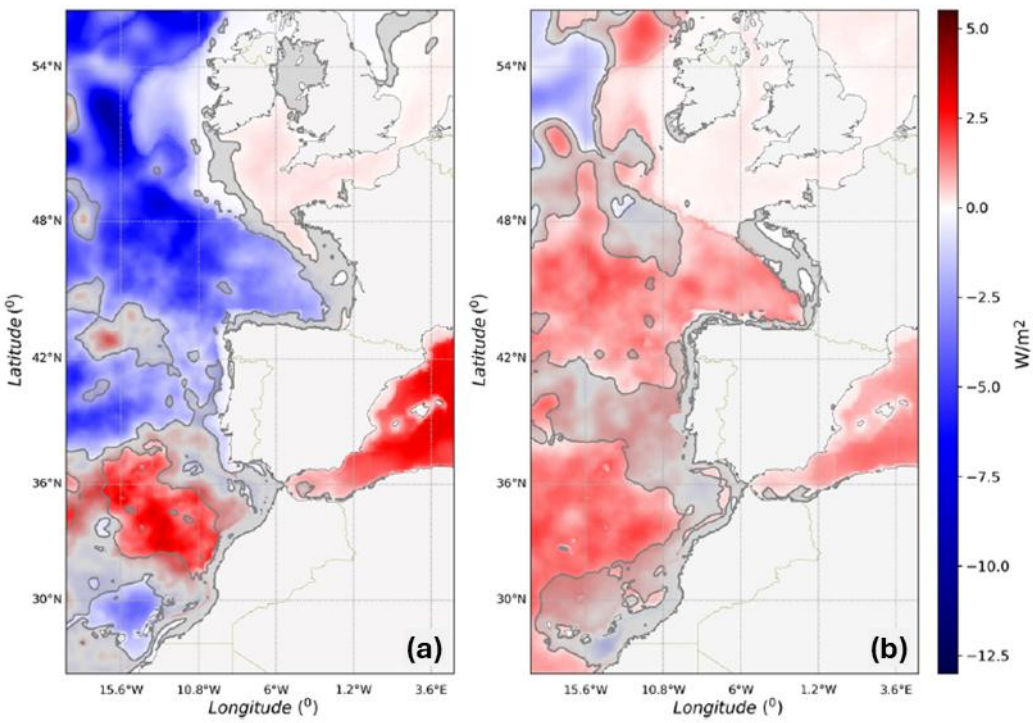

Figure 3: Regional maps of OHC trend computed at 0–2000m over the periods 2005–2019 (a) and 1993–2021 (b). Trends computed using four Copernicus Marine products (IBI-REA, GLO-REA, CORA, and ARMOR), providing a multi-product estimation. Shaded colours represent mean trends (using all products), while shaded grey indicates areas with less robust signatures (where

the noise – ensemble standard deviation of trends – exceeds the signal – ensemble mean trend) (figure obtained from de Pascual Collar et al., 2023).

3) REFERENCES

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